

- 6** An object of mass 1.5 kg is sliding with a velocity of 3.0 m s^{-1} on a smooth horizontal surface towards another stationary object of mass 2.0 kg. The resulting head-on collision is completely inelastic.

If the duration of the collision is 0.050 s, what is the average force exerted between the objects during the collision?

- A** 39 N
- B** 51 N
- C** 90 N
- D** 120 N

7 For a spring the relationship between the force F applied to it and the extension x of the